

# 1 Cloud Computing Intro

- Clusters - Many locally connected computers
- Grids - Loosely coupled and widely distributed clusters or computers
- Clouds - IT resources as a utility
- Edge/Fog - Cloud services in closer proximity to users and edge devices

## 1.1 Clusters

Mostly homogenous commodity compute resources and software stacks viewed as one system. Interconnected by a low latency and high-bandwidth LAN.

Goal is to improve availability, resource utilisation, and price/performance.

## 1.2 Grids

Widely distributed heterogenous compute resources, connected via a slow network, i.e. the internet. Heterogenous software stacks unified by a middleware used for a common goal.

## 1.3 Clouds

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

This cloud model is composed of five essential characteristics, three service models, and four deployment models

(National Institute of Standards and Technology(NIST))

## 1.4 NIST Definition's Dimensions

### 1.4.1 Five Characteristics

- On-demand self-service: No human interaction required for resource provisioning.
- Broad network access: Accessible over networks with standard mechanisms.
- Resource Pooling: Pooled resources dynamically shared among several consumers, with location independence.
- Rapid elasticity: Capabilities can be provisioned/released on demand.
- Measured service: Resource usage is monitored, controlled, and reported.

### 1.4.2 Three Service Models

- Software as a Service (SaaS) : Providers application runs on cloud infrastructure and can be accessed over the network, consumer does not control/manage underlying infrastructure.
- Platform as a Service (PaaS) : Consumer can deploy application onto cloud infrastructure using programming languages, libraries, and tools supported by the provider. Consumer again does not control/manage underlying infrastructure infrastructure.
- Infrastructure as a Service (IaaS) : Provider provides processing, storage, and network resources to consumer.

Consumer has control over OS, storage, and deployed apps, but does not **control** underlying infrastructure.

### 1.4.3 Four Deployment Models

|                                       | Private Cloud                                  | Community Cloud                                 | Public Cloud                                                     | Hybrid Cloud                                                                                                                                                      |
|---------------------------------------|------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User of the cloud infrastructure?     | Single organization                            | Organizations with shared concerns              | Open for the general public                                      | Composition of private / community / public cloud:                                                                                                                |
| Owner of the cloud infrastructure?    | Organization, third party, combination thereof | Organizations, third party, combination thereof | Business, academic, government organization, combination thereof | <ul style="list-style-type: none"> <li>• Clouds remain distinct, but bound together by standard mechanisms</li> <li>• General goal: Enable portability</li> </ul> |
| Location of the cloud infrastructure? | On premise, off premise                        | On premise, off premise                         | On premise of cloud provider                                     |                                                                                                                                                                   |

# 2 Virtualisation, VMs, and Containers

## 2.1 Challenges for Cloud Providers

- Rapid Provisioning - resources must be available to the consumer quickly, and with no human interaction needed during provisioning.
- Elasticity - create illusions of infinite resources, while managing data center in a cost-efficient manner.
- Isolation of different consumers - users should not interfere with each other
- Performance - maintain good performance despite these(or other) challenges.

## 2.2 Virtualisation

The main idea on the IaaS level is to provide *virtual resources*, i.e. compute resources in the form of virtual machines and containers. Can also include other computing resources like virtual networks/network functions, or virtual storage,

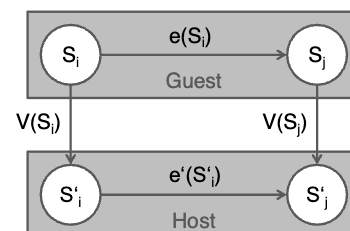
The NIST definition of virtualisation is

Virtualisation is the **simulation of the software and/or hardware upon which other software runs**. This simulated environment is called a virtual machine (VM) (NIST)

Virtualisation can transform a real system so that it looks like a different virtual system or like multiple virtual systems. The real system is referred to as the host, and the virtual system(s) is referred to as a guest.

### 2.2.1 Formal Definition

Isomorphism  $V$  with states of machines  $S_i, S_j$  and a sequence of operations  $e$ .



Isomorphism  $V$  maps guest state to host state such that for  $e$  that modifies the guest's state from  $S_i$  to  $S_j$  there exists

a corresponding sequence of operations  $e'$  that performs an equivalent modification between the hosts states  $S_i'$  to  $S_j'$

## 2.3 Virtual Machines

VM: hardware that is used by the operating system and running process is not real, but virtual. OS and processes run on software emulating software.

VMs can have different hardware characteristics (including differences from host), and are often started from available images with an OS and some applications installed

### 2.3.1 Use cases

- Running a different OS than installed on host
- Operating multiple isolated environments on host
- Pool of resources shared by multiple users and applications in a private cloud

## 2.4 Containers

Lightweight OS level virtualisation, virtual environments for a single application, processes run in isolation *within an OS*. Containers rely on containment features of the OS kernel.

Used for isolating applications on the same host from each other, and for bundling dependencies and the environment of application to ship software.

Containers can have different sizes, are usually started from images, and can be run on cloud services

### 2.4.1 Containers vs VMs

Containers have no virtual hardware, cannot run an application build for a different architecture or OS kernel. They have reduced scope and isolation, as containerised applications share the same kernel and are only isolated on process level.

VMs have larger images, slower startup, and more overhead per VM on a physical node.

## 3 Example Qs

### 3.1 LT

#### 3.1.1 Quiz 1

**Q1: Which of the following statements correctly differentiates between clusters, clouds, and grids in computing?**

e. Clusters locally connect computers to form a single system, grids integrate widely distributed systems for common tasks, and clouds offer scalable computing resources as a service.

**Q2: Which of the following best matches the cloud service model with its primary function?**

f. SaaS (Software as a Service): Delivers complete software solutions accessible via the Internet (e.g. as Web applications).

**Q3: Which of the following is NOT one of the five essential characteristics of cloud computing as defined by NIST?**

f. On-Demand Self-Service

g. Broad Network Access:

h. **High Availability**

i. Resource Pooling

j. Rapid Elasticity

**Q4: Which of the following best describes virtualization?**e.

Virtualization allows multiple virtual instances to run on a single physical hardware resource, abstracting and sharing resources.

**Q5: Which of the following statements correctly describes the use cases of VMs versus containers?** d. VMs allow running applications requiring a different operating system, while containers are more suitable for shipping software components with their dependencies.

#### 3.1.2 Quiz 2

**Q1: Which of the following statements correctly differentiates between Type 1 and Type 2 hypervisors in terms of overhead and use cases?**

Type 1 hypervisors run on bare-metal hardware, which minimizes overhead, making them ideal for clouds, whereas Type 2 hypervisors run on top of an operating system, which itself needs resources, but they enable virtualization on top of a personal computing environment.

**Q2: Which computing resources are typically virtualized to allow multiple virtual machines to operate on a single physical host?**

Access to CPUs, memory, I/O devices, and storage devices all need to be managed to create isolated environments for virtual machines on a shared host system.

**Q3: Which of the following statements accurately describes Popek and Goldberg's theorem for efficient virtualization and its implications for x86 processors?**

The theorem requires that all privileged and sensitive instructions trap to the hypervisor when executed in user mode. x86 processors did not meet this requirement, requiring binary re-writing.

**Q4: In the context of full virtualization, what is true for shadow page tables?**

Shadow page tables, which map virtual memory addresses as used by the guest OS directly to the actual physical memory of the host, are maintained by the hypervisor and used by hardware.

**Q5: In Xen, how is I/O virtualized to allow guest virtual machines isolated access to physical devices?**

Xen uses a split-driver model where lightweight virtual device drivers in guest virtual machines communicate with Xen through hypercalls and events, and Xen, in turn, uses the actual physical device drivers that are part of the privileged domain (Dom0).

#### 3.1.3 Quiz 3

**Q1: Which of the following Linux kernel features is not used for OS-level virtualization?**

a. chroot

b. namespaces

c. **cron**

d. cgroups

**Q2: Which of the following is not a feature that Docker provides on top of a container library (such as libcontainer)?**

a. Image distribution

b. Build tools

c. Orchestration

d. **Fault tolerance and auto-scaling**

**Q3: When comparing the performance overhead of virtual machines (VMs) and containers, which of the following statements is true?**

Using hardware devices through virtual devices and virtual device drivers can lead to higher latencies and latency variations in VMs.

**Q4: Which of the following is not an advantage of microservice architectures?** a. Microservices enable independent deployment and scaling of services. b. Microservices can improve resilience by isolating failures within individual services. c. Microservices allow flexibility in choosing different technologies for different components. d. **Microservices increase cluster resource utilization by running more services simultaneously.**

**Q5: Which of the following statements is true for features of Kubernetes?**

Kubernetes probes nodes and pods in a configurable interval, noticing failures with a delay as large as the interval.

#### 3.1.4 Quiz 4

**Q1: Why do cloud operating systems like OpenStack not typically use different physical hosts for different host roles (such as compute hosts, storage hosts, network hosts, and controller hosts)?**

To reduce costs by spreading out storage, compute, and networking across clusters.

**Q2: Which of the following statements on virtual networking is not correct?**

b. Virtual Network Functions (VNFs) refer to the software-based network appliances like firewalls, load balancers, and routers running on physical infrastructure to perform traditional networking tasks.

**Q3: Which of the following is not a correct step or feature of live VM migration in Xen?**

Xen synchronizes the source and destination VMs by executing the same CPU instructions in real time during live migration.

**Q4: Which of the following issues is not effectively addressed by adopting the Infrastructure-as- Code (IaC) paradigm?**

Resource underprovisioning: IaC can mitigate resource bottlenecks by adequately scaling the infrastructure for the code.

**Q5: Which of the following types of testing and analysis is not commonly performed during canary deployments to ensure system stability and performance before a full rollout?**

Chaos Testing: Randomly introducing controlled failures to the system to evaluate the new deployment's resiliency.

#### 3.1.5 Quiz 5

**Q1: Which of the following is not a correct category of computer system lifecycle emissions?**

Development emissions: emissions from design, development, and testing of software.

**Q2: Which of the following statements is not true for the power consumption of computing hardware?**

As the peak performance increases from one server generation to the next, it becomes more and more important to utilize server hardware well for energy-proportional computing

**Q3: Which of the following statements about Power Usage Effectiveness (PUE) is not true?**

PUE is often better for small, specialized data centers, as they need less energy overall.

**Q4: What is not a reason why operational cloud carbon footprint assessments are difficult?**

Grid carbon intensity varies between regions, seasons, weekdays, and times of day.

**Q5: What is not a reason why carbon-aware computing is not used more in practice?**

Limited applicability: The majority of cloud applications are not delay-tolerant but latency- critical, user-facing Web applications, and those cannot be managed based on varying grid carbon intensity.

## 3.2 YE

### 3.2.1 Quiz 1

**What is the primary difference between reliability and availability in cloud systems?**

The correct answer is: Reliability measures the system's ability to perform correctly, while availability measures the proportion of time the system is operational

**Which of the following is a common data partitioning scheme used in cloud databases?**

The correct answer is: Hash partitioning

**What does MTTF stand for in reliability engineering?**

The correct answer is: Mean Time To Failure

**Which statement correctly describes the difference between "scaling up" and "scaling out" in cloud computing?**

The correct answer is: Scaling up means adding more powerful resources to an existing server, while scaling out means adding more servers

**In the context of cloud systems, what is the relationship between a fault and a failure?**

The correct answer is: A fault is a defect that may lead to a failure

**Redundancy by replication primarily aims to improve**

The correct answer is: Availability

**In which failover strategy is only one server actively handling requests while the other server remains on standby?**

The correct answer is: Active-Passive

**What does "System Function" refer to?**

The correct answer is: The intended purpose of the system

**What is the primary purpose of implementing the Circuit Breaker pattern in cloud-based systems?**

The correct answer is: To prevent cascading failures by stopping calls to failing services

**Which of the following is a key benefit of an Active-Active failover strategy?**

The correct answer is: Short time to switch between active servers

### 3.2.2 Quiz 2

**What is the primary purpose of the Data Center Manageability Interface (DCMI)?**

The correct answer is: To provide standardised interfaces for monitoring and managing power consumption and thermal conditions

**Which of the following statements about the Intelligent Platform Management Interface (IPMI) is CORRECT?**

The correct answer is: IPMI allows administrators to monitor and manage servers even when the operating system is not running or the server is powered off

**Which TWO of the following are NOT standard hardware redundancy practices in cloud data centres?**

The correct answers are: Requiring manual administrator intervention to activate backup systems during hardware failures, Requiring all applications to maintain their own redundancy

**Which statement about Life Cycle Assessment (LCA) methodologies is INCORRECT?**

The correct answer is: LCA is split into 3 scopes: direct, indirect, and hidden

**A healthcare company wants to process patient data while ensuring privacy and minimal latency. Which architecture should you recommend?**

The correct answer is: Edge computing with local processing

**Which cloud deployment model offers the greatest level of control to the customer?**

The correct answer is: Private cloud

**Which cloud provisioning level is best described as providing managed runtime environments where users deploy applications without managing underlying infrastructure?**

The correct answer is: PaaS

**Which of the following best describes common PaaS pricing models?**

The correct answer is: PaaS uses consumption-based pricing that includes both infrastructure resources and platform management costs

**Which statement best describes the relationship between fog computing and edge computing?**

The correct answer is: Fog acts as an intermediate layer between the edge and the cloud

**When would edge computing be most beneficial?**

The correct answer is: For real-time processing with low latency

## API Gateways – OnlyScran [5 mins]

- o Delays in processing payments at surge times causes the entire system to slow down, affecting even basic menu browsing.
- o Group discussion points:
  1. How could a slow payment service impact menu browsing?
  2. Design a timeout solution, considering placement, duration, and failure handling.
  3. What metrics would validate your solution?
- 1. *Payment service likely shares resources with menu and other services*
  - e.g. blocked database connections, depleted thread pool
- 2. *Timeout solution*
  - Place timeouts at API gateway; e.g. 3 seconds for payment service requests
  - Add circuit breaker pattern; e.g. after 5 payment timeouts, route to backup payment processor
  - Implement graceful degradation; allow menu browsing even if payments are down
  - Cache menu data to reduce load on contented resources such as the database
- 3. *Request (payment) processing latency, #timeouts, resource utilization (CPU, mem, DB conns), user abandonment frequency*

## API Gateways – OnlyScran II [7 mins]

- o How would you leverage cloud-native services to improve the resilience and scalability of the payment processing component?
- o *Use a **message queue** to decouple the order placement service from the payment processing service. When an order is placed, a message is sent to the queue, and the payment processing service consumes messages from the queue asynchronously. This prevents the order placement process from being blocked by slow payment processing.*
- o ***Serverless functions** can be used to handle payment processing. This allows for automatic scaling and reduces the operational overhead of managing servers.*

## Failover Strategies – Exercise [8 mins]

- o Match each scenario with the most appropriate failover strategy (Active-Active / Active-Passive). Provide justification.

| E-commerce Product Catalogue                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Content Delivery Network (CDN) Cache                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Online Banking Authentication Service                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Characteristics:</b> <ul style="list-style-type: none"><li>• Mainly serves product info and images</li><li>• Read-heavy workload</li><li>• Data can be eventually consistent (can handle temporary inconsistencies)</li></ul> <b>Best Strategy: Active-active</b><br><b>Justification:</b> <ul style="list-style-type: none"><li>• High read throughput can be distributed</li><li>• Perfect for horizontal scaling</li><li>• Improved global performance through distribution</li><li>• Updates are infrequent and can be propagated</li></ul> | <b>Characteristics:</b> <ul style="list-style-type: none"><li>• Static content distribution</li><li>• Regional performance requirements</li><li>• Read-only copies</li><li>• Can handle stale data briefly</li></ul> <b>Best Strategy: Active-active</b><br><b>Justification:</b> <ul style="list-style-type: none"><li>• Multiple active nodes improve performance</li><li>• Content replication is straightforward</li><li>• No write conflicts to manage</li><li>• Better user experience through locality</li></ul> | <b>Characteristics:</b> <ul style="list-style-type: none"><li>• Session state management</li><li>• Security-critical operations</li><li>• Audit requirements</li></ul> <b>Best Strategy: Active-passive</b><br><b>Justification:</b> <ul style="list-style-type: none"><li>• Centralized session management</li><li>• Reduced attack surface</li><li>• Simpler security auditing</li><li>• Clear authority for authentication</li></ul> |